

Yash Bajpai

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EDUCATION

Vellore Institute of Technology (VIT)

B.Tech. Information Technology; CGPA: 8.75 / 10

Vellore, India

2023 – 2027

EXPERIENCE

Software Engineering Intern

2026

Maruti Suzuki India Limited

On-Site

- Engineered an automated Excel-to-MongoDB ETL pipeline processing 10,000+ manufacturing alarm records through REST APIs, powering a full-stack Industrial Alarm Analytics platform using Node.js, Express.js, MongoDB and JavaScript.
- Designed a polyglot data architecture combining MongoDB for alarm/event data with PostgreSQL for relational manufacturing metadata, implementing schema relationships, foreign keys and SQL-based analytical queries.
- Built aggregation-driven KPI dashboards with filtering, pagination and Pareto analysis for production downtime, and implemented an Isolation Forest anomaly-detection model to identify abnormal machine alarm patterns and surface potential downtime risks.

Software Engineering Intern

2026

AI Blocks

Remote

- Madras High Court Cause List Parser (DAGP) — Built a large-scale document data-extraction pipeline handling 4,000+ cases per PDF using Document Anchor Graph Parsing, spatial tokenization, and graph segmentation; reached 97–99% extraction accuracy at 2–4 sec/PDF with zero OCR or LLM dependencies.
- Designed and implemented modern frontend interfaces for client projects, emphasizing responsive design, performance, and user experience.

Data Engineering Intern

2025

Algoleap

Remote

- Authored a modular Python/BeautifulSoup data-extraction pipeline targeting Glassdoor and AmbitionBox with site-specific parsing logic, sustaining a 95% parse success rate.
- Applied NLP-based sentiment analysis to isolate negative reviews; exported validated, structured CSV/JSON output with field-level schema enforcement for downstream analysis.
- Instrumented exponential-backoff retries, rotating proxy support, and structured logging.

TECHNICAL PROJECTS

Stress Intelligence Platform | PySpark, Spark SQL, Parquet, scikit-learn

2026

- Designed a distributed data-processing pipeline handling 130K+ weekly student records with columnar Parquet partitioning, reducing analytical query latency by 60% over row-based storage.
- Engineered 12+ temporal entropy features (Shannon entropy, drift, acceleration, volatility) and trained a class-balanced Random Forest classifier achieving 88% recall on high-risk cases — a 3× lift over the naive baseline.
- Shipped an incremental inference engine producing student-level intervention alerts in under 1 second from live input streams, enabling near real-time risk monitoring.

DocGraph AI | Full-Stack GraphRAG Document Intelligence

2026

- Created a hybrid RAG system combining dense HNSW vector search with semantic graph traversals over Neo4j (GraphRAG), and orchestrated an asynchronous multi-agent FastAPI backend migrated to the Gemini Batch API to eliminate OOMs and rate-limit errors.
- Designed a write-time entity resolution engine using inline HNSW vector blocking with a 3-way similarity decision boundary to canonicalize entities across thousands of documents and prevent graph fragmentation.

AEGIS-ZONE | Zero-Trust Workstation Security System | FastAPI, Next.js, React, ESP32, WebSockets

2026

- Developed a full-stack Zero-Trust workstation prototype using FastAPI, Next.js, React, ESP32, and WebSockets, integrating biometric authentication and wireless intrusion detection for continuous access verification.
- Implemented real-time AES session key generation/invalidation using SHA3-256, live dashboard monitoring, and automated workstation lockdown upon biometric failure or unauthorized wireless threats.

TECHNICAL SKILLS

Languages: Python, C/C++, JavaScript (ES6+), SQL, HTML5, CSS3

Data Science & ML: PySpark, YOLOv8, CNN, OpenCV, NumPy, SciPy, Pandas, scikit-learn, Feature Engineering, Random Forest, Model Evaluation

Data Engineering & Databases: MongoDB, DynamoDB, MySQL, ETL/Data Pipelines, KPI/Pareto Analysis

Cloud & DevOps: AWS (S3, CloudFront, Elastic Beanstalk, DynamoDB), OAuth 2.0

Web: React.js, RESTful APIs, JWT, FastAPI

Core CS: Data Structures & Algorithms, Software Engineering, OOP, Java